CIS41032: Practical for Advance Database Management System

Department of ICT Faculty of Technology
South Eastern University of Sri Lanka

Lab Sheet 04 & 05

Title: Update, Delete Commands in MongoDB.

Objective:

- Understand about Update operations.
- Understand delete commands.

Delete Operations

1. Delete students who have the status "**Graduated**".
2. Remove students from **Grade 11** who enrolled **before 2020**.
3. Delete a specific student record for "**James White**".
4. Remove the "**Scholarship**" field from all students.

Exercise 01

Dataset: Books Collection

BookID	Title	Author	Genre	Published Year	Copies Available	Total Copies	Status
301	The Great Gatsby	F. Scott Fitzgerald	Fiction	1925	3	5	Available
302	To Kill a Mockingbird	Harper Lee	Fiction	1960	1	4	Available
303	1984	George Orwell	Dystopian	1949	0	6	Borrowed
304	The Catcher in the Rye	J.D. Salinger	Fiction	1951	2	3	Available
305	Moby-Dick	Herman Melville	Adventure	1851	5	5	Available
306	Pride and Prejudice	Jane Austen	Romance	1813	1	4	Available
307	The Hobbit	J.R.R. Tolkien	Fantasy	1937	0	7	Borrowed

308	War and Peace	Leo Tolstoy	Historical	1869	2	5	Available
309	Hamlet	William Shakespeare	Drama	1603	4	6	Available
310	Crime and Punishment	Fyodor Dostoevsky	Crime	1866	3	4	Available

1. Create a new database called **LibraryDB.**
2. Create a collection named **Books**.
3. Insert all the book records from the dataset above.
4. Increase the total copies of "**1984**" by **2** and adjust available copies accordingly.
5. Mark "**Pride and Prejudice**" as "**Reserved" if copies available are less than 2.**
6. Update all books published **before 1900** to have a **new field "Classic**": true.
7. Change the **genre of "The Hobbit**" to "**Epic Fantasy**".
8. Add a "**LastBorrowedDate**" field to books that are currently borrowed (set a sample date).
9. Delete all books where **TotalCopies** is **less than 3** (outdated books removal).
10. Remove the "**Crime and Punishment**" book from the database.
11. Remove the "**Classic"** field from books **published after 1900** (to correct the dataset).
12. Delete books that have been **borrowed but have 0 total copies left**.

<u>Exercise 02</u>

ProductID	**Name**	**Category**	**Price (USD)**	**Stock Quantity**	**Supplier**	**ExpiryDate**	**Discount (%)**	**Availability**
101	Apple	Fruits	1.50	100	Fresh Farms	2025-06-20	5	In Stock
102	Orange Juice	Beverages	3.00	50	Nature's Best	2024-12-10	10	In Stock
103	Rice (5kg)	Grains	12.00	40	Agro Supplies	2026-01-01	0	In Stock
104	Chicken Breast	Meat	8.50	20	Farm Fresh	2024-11-15	15	In Stock

105	Whole Wheat Bread	Bakery	2.50	60	Baked Goods	2024-09-05	5	In Stock
106	Milk (1L)	Dairy	2.00	80	Dairy Fresh	2024-08-20	0	In Stock
107	Olive Oil (1L)	Cooking Oil	10.00	30	Health Foods	2026-05-15	5	In Stock
108	Chocolate Bar	Snacks	1.20	200	Sweet Treats	2025-02-18	20	In Stock
109	Cereal Box	Breakfast	4.50	90	Healthy Start	2025-03-22	10	In Stock
110	Eggs (Dozen)	Dairy	3.50	70	Farm Fresh	2024-07-30	0	In Stock

1. Create a new database called SupermarketDB.
2. Create a collection named Products.
3. Insert all product records into the Products collection.
4. Retrieve all products in the collection.
5. Find details of the product named "Olive Oil (1L)".
6. Find all products in the "Dairy" category.
7. Find products that are priced above **$5.00**.
8. Find products that have a discount greater than **10%**.
9. Find products with stock quantity below **50**.
10. Find all products expiring before **December 31, 2024**.
11. Retrieve all products supplied by "Farm Fresh".
12. Find products where the name starts with "C".
13. Increase the price of "Apple" by **$0.50**.
14. Update all products in the "Bakery" category to include "Gluten-Free": false.
15. Set "Availability": "Out of Stock" for products with stock quantity less than **20**.
16. Apply an extra **5% discount** to products in the "Snacks" category.
17. Change the supplier of "Milk (1L)" to "Dairy Best".
18. Add a "Rating" field with a default value of **4.0** for all products.

19. Remove all products that have expired (ExpiryDate before today).
20. Delete the "Chocolate Bar" product from the collection.
21. Remove the "Discount" field from all products in the "Grains" category.